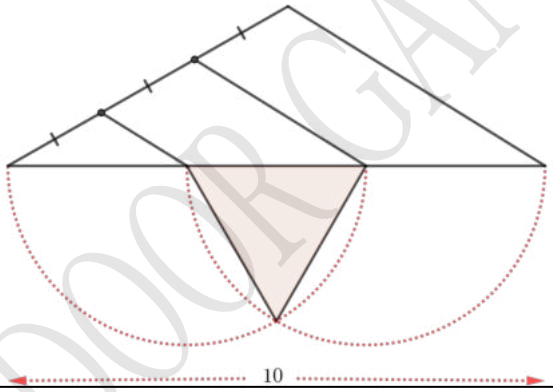
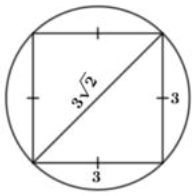
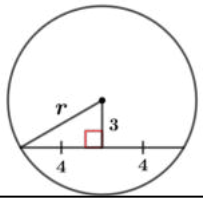
A

SUMMATIVE ASSESSMENT 2025 - 26

MATHEMATICS – ANSWER KEY

E-903

Class : IX

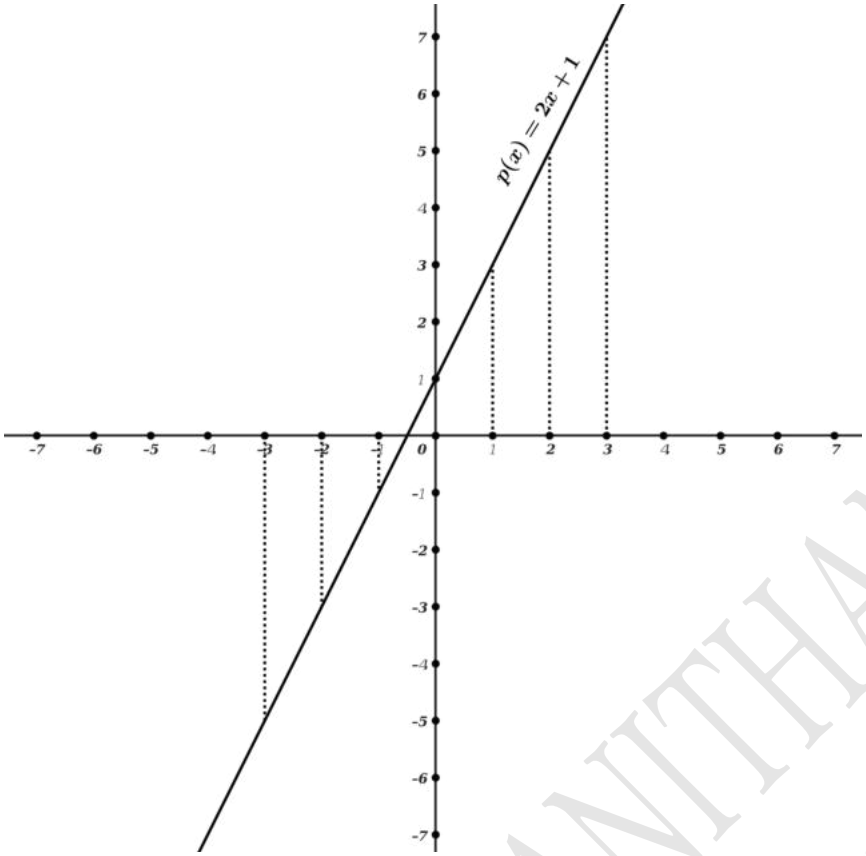
Qn no	Key	Score
	SECTION A	
1	A. $\sqrt{3}$	1
2	C. 2 $\left[\frac{-1+5}{2} = \frac{4}{2}\right]$	1
3	C. $\frac{1}{4}$ $\left[\frac{1}{2} \times \frac{1}{2}\right]$	1
4	B. 20 centimetres $[\,30 : 120 = 1 : 4\,]$	1
5	B. 21 $\left[\frac{18+24+21+20+22}{5} = \frac{105}{5}\right]$	1
6	A. (i) and (iv)	1
7	C. Both the Statements are two , Statement II is the reason of Statement I	1
8	D. Both the Statements are two , Statement II is not the reason of Statement I	1
	SECTION B	
9	For drawing a line of length 10 cm and dividing it into three equal parts For completing the equilateral triangle 	2 1
10A	(i) For identifying that the diagonal of the square is the diameter of the circle Radius of the circle $= \frac{3\sqrt{2}}{2}\ cm$ (ii) Perimeter of the circle $= 2\pi \times \frac{3\sqrt{2}}{2} = 3\sqrt{2}\pi\ cm$ 	1 1 1
10B	$r^2 = 3^2 + 4^2$ $r = \sqrt{25} = 5\ cm$ Area of the circle $= \pi \times 5^2 = 25\pi\ sq.cm$ 	1 1 1
11	Price of 4 pencils $= 59 - 39 = 20$ (i) Price of a pencil $= \frac{20}{4} = 5\ Rs$ (ii) Price of a pen $= \frac{39-15}{2} = \frac{24}{2} = 12\ Rs$ OR	1 1 1

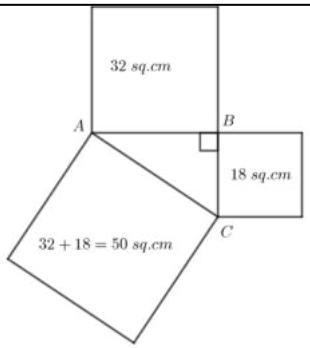
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	If we take the price of a pencil is x Rs and the price of a pen is y Rs, then $2x + 3y = 39 \quad (1)$ $2x + 7y = 59 \quad (2)$ $(2) - (1) \Rightarrow 4y = 20 \Rightarrow y = \frac{20}{4} = 5\ Rs$ Put $y = 5$ in equation (1) , we get $2x + (3 \times 5) = 39 \Rightarrow 2x + 15 = 39 \Rightarrow 2x = 24 \Rightarrow x = 12\ Rs$	
12A	(i) If we take the number of sides of the regular polygon is n, Measure of an outer angle $= \frac{360}{n} = 360 \times \frac{1}{n}$ Measure of an outer angle is inversely proportional to the number of sides. (ii) Proportionality constant $= 360$	1 1 1
12B	(i) Distance travelled in one rotation = Circumference of the wheel $= 2\pi \times 50 = 100\pi\ cm$ (ii) Distance travelled is proportional to the number of rotations. Proportionality constants $= 100\pi$ [Distance travelled = Number of rotations x Circumference of the wheel]	1 1 1
13	(i) Base area of the cylinder $= \pi \times 7^2 = 49\pi\ sq.cm$ (ii) volume of the cylinder $= 49\pi \times 15 = 735\pi\ cubic.cm$	1 2
14	Circumference of the circle $= 2\pi \times 5 = 10\pi\ cm$ Central angle of the arc $= \frac{6\pi}{10\pi} \times 360$ $= 216^0$ OR If we take the central angle of the arc is x^0, $\frac{x}{360} \times 2\pi \times 5 = 6\pi$ $\frac{x}{360} \times 10\pi = 6\pi \Rightarrow \frac{x}{36} = 6 \quad x = 6 \times 36 = 216^0$	1 1 1
	SECTION C	
15	For showing how p(x) changes with x using a table. For marking x's on the horizontal line For showing positive values of p(x) by drawing perpendiculars upward and negative values of p(x) by drawing perpendiculars downwards. For joining the ends of the perpendiculars.	1 1 1 1

x	-3	-2	-1	0	1	2	3
$p(x)$	$-6+1=-5$	$-4+1=-3$	$-2+1=-1$	$0+1=1$	$2+1=3$	$4+1=5$	$6+1=7$

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16A	(i) $AB = \sqrt{32} = \sqrt{16 \times 2} = 4\sqrt{2}\ cm$ $BC = \sqrt{18} = \sqrt{9 \times 2} = 3\sqrt{2}\ cm$ $AC = \sqrt{50} = \sqrt{25 \times 2} = 5\sqrt{2}\ cm$ (ii) Perimeter of the triangle ABC $= \sqrt{32} + \sqrt{18} + \sqrt{50}$ $= 12\sqrt{2}\ cm$ [Let's do like this also, $AC = \sqrt{AB^2 + BC^2} = \sqrt{32 + 18} = \sqrt{50}$]  	1 1 1
16B	(i) Perimeter of the triangle $= 13 + 14 + 15 = 42$ cm (ii) $s = \frac{1}{2} \times 42 = 21\ cm$ Area of the triangle $= \sqrt{21 \times (21-13) \times (21-14) \times (21-15)}$ $= \sqrt{21 \times 8 \times 7 \times 6}$ $= \sqrt{(7 \times 3) \times (2 \times 2 \times 2) \times 7 \times (2 \times 3)}$ $= 7 \times 3 \times 2 \times 2 = 84\ sq.cm$	1 1 1 1
17	(i) Volume of the rectangular prism $= 5 \times 8 \times 16 = 640\ cubic.cm$ (ii) volume of a cube $= 4 \times 4 \times 4 = 64\ cubic.cm$ Number of cubes $= \frac{640}{64} = 10$	1 1 2
18	(i) $p(0) = 4$ $[\ p(x) = x^2 - 5x + 4\ ]$ $p(1) = 1 - 5 + 4 = 0$ $p(-2) = (-2)^2 - [5 \times (-2)] + 4 = 4 + 10 + 4 = 18$ (ii) $p(1) = 3 + 2 + 1 = 6$ $[\ p(x) = 3x^2 + 2x + 1\ ]$ $p(10) = 3 \times 10^2 + (2 \times 10) + 1 = 321$	1 1 1

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19

Height	Number of children	Total height
140	5	140 x 5 = 700
142	10	142 x 10 = 1420
145	8	145 x 8 = 1160
148	5	148 x 5 = 740
150	2	150 x 2 = 300
Total	30	4320

Mean height $= \frac{4320}{30} = 144\ cm$ — 3, 1

20A

(i) For recognising that the given triangle is equilateral — 1

Central angle of a sector $= 360 - 60 = 300^0$ — 1

(ii) Area of the shaded portion $= 3 \times$ Area of a sector

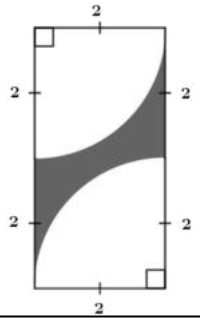
$= 3 \times \frac{300}{360} \times \pi \times 3^2 = \frac{45\pi}{2}\ sq.cm$ — 2

20B

(i) Arc length of the sector $= \frac{90}{360} \times 2\pi \times 2 = \pi\ cm$ — 2

(ii) Perimeter of the shaded part $= \pi + 2 + \pi + 2$

$= 2\pi + 4\ cm$ — 2

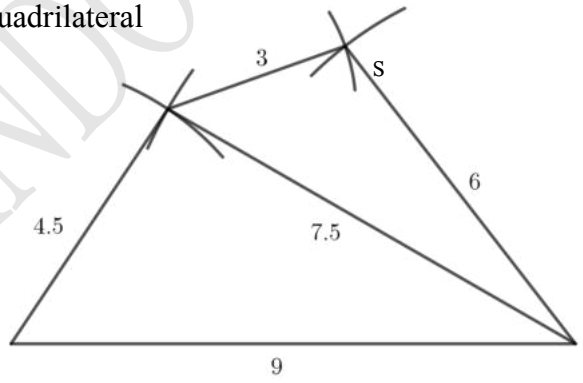


SECTION D

21

For calculating $1\frac{1}{2}$ times of the sides and the diagonal of the quadrilateral — 2

For completing the quadrilateral — 3



$6 \dashrightarrow 6 \times 1.5 = 9$
$3 \dashrightarrow 3 \times 1.5 = 4.5$
$5 \dashrightarrow 5 \times 1.5 = 7.5$
$4 \dashrightarrow 4 \times 1.5 = 6$
$2 \dashrightarrow 2 \times 1.5 = 3$

22A

(i) Equilateral triangular prism OR Triangular prism — 1

(ii) Base area of the prism $= \frac{\sqrt{3}}{4} \times 4^2 = 4\sqrt{3}\ sq.cm$ — 1

(iii) Lateral surface area of the prism = Base perimeter x height $= 3 \times 4 \times 5 = 60\ sq.cm$ — 1

Total surface area of the prism = 2 x Base area + Lateral surface area — 1

$= 2 \times 4\sqrt{3} + 60 = 8\sqrt{3} + 60\ sq.cm$

Total cost $= (8\sqrt{3} + 60) \times 80 = 640\sqrt{3} + 4800 = (640 \times 1.73) + 4800 = 5907.20\ Rs$ — 1

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22B

(i) Base radius of the tank $= \frac{120\pi}{2\pi} = 60\ cm$ — 1

(ii) Base area of the tank $= \pi \times 60^2 = 3600\pi\ sq.cm$ — 1

(iii) Volume of the tank = Base area x height $= 3600\pi \times 150 = 540000\pi\ cubic.cm$ — 1

$= \frac{540000\pi}{1000} = 540\pi\ litres$ — 1

$= 540 \times 3.14 = 1695.60\ litres$ — 1

23

$p(x) = ax^2 + bx + c$

(i) $p(0) = 2$, $p(1) = 0$, $p(2) = 0$ — 1

(ii) $c = 2$ — 1

$a + b = -2 \quad (1)$

$4a + 2b = -2 \implies 2a + b = -1 \quad (2)$ — 1

$(2) - (1) \implies 2a - a + b - b = -1 - (-2) \implies a = 1$

$b = -2 - 1 = -3$ — 1

$\therefore\ p(x) = x^2 - 3x + 2$ — 1

24A

(i) Sum of the angles of the regular hexagon $= 4$ x $180 = 720^0$ — 1

Central angle of an arc $= \frac{720}{6} = 120^0$ — 1

(ii) We get two circles of radius 2 cm by joining these arcs.

Perimeter of the shaded part = 2 x Circumference of a circle — 1

$= 2 \times 2\pi \times 2 = 8\pi\ cm$ — 1

(iii) Area of the shaded part = Area of the regular hexagon – 2 x Area of a circle

$= \frac{6\sqrt{3}}{4} \times 4^2 - 2 \times \pi \times 2^2 = 24\sqrt{3} - 8\pi\ sq.cm$

OR

(ii) Perimeter of the shaded part = 6 x Arc length of a sector — 1

$= 6 \times \frac{120}{360} \times 2\pi \times 2 = 8\pi\ cm$.

(iii) Area of the shaded part = Area of the regular hexagon – 6 x Area of a sector

$= \frac{6\sqrt{3}}{4} \times 4^2 - 6 \times \frac{120}{360} \times \pi \times 2^2 = 24\sqrt{3} - 8\pi\ sq.cm$

24B

(i) Length of an arc $= \frac{12\pi}{3} = 4\pi\ cm$ — 1

(ii) Circumference of the circle $= 4\pi\ cm$ — 1

Radius of the circle $= \frac{4\pi}{2\pi} = 2\ cm$ $[\ 2\pi r = 4\pi\ ]$ — 1

(iii) Area of the circle $= \pi \times 2^2 = 4\pi$ — 1

$= 4 \times 3.14 = 12.56\ sq.cm$ — 1

25

Daily wages	Number of workers	Class wage	Total wage
550 – 650	10	$\frac{550 + 650}{2} = 600$	600 x 10 = 6000
650 – 750	13	$\frac{650 + 750}{2} = 700$	700 x 13 = 9100

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		3
	(i) 50	1
	(ii) Mean daily wage $= \frac{39200}{50} = 784\ Rs$	1
26A	(i) (a) $4 \times \frac{\pi}{3}$	1
	(b) $10 \times \frac{\pi}{3}$	1
	(ii) $x \times \frac{\pi}{3}$	1
	(iii) Arc length is proportional the radius.	1
	Proportionality constant $= \frac{\pi}{3}$	1
26B	(i) (a) $\frac{100}{20} = 5$	1
		1
	(b) $\frac{100}{50} = 2$	1
	(ii) $y = \frac{100}{x}$	1
	(iii) Speed is inversely proportional to the time.	1
	(iv) Proportionality constant = 100	1

750 – 850	9	$\frac{750 + 850}{2} = 800$	800 x 9 = 7200
850 – 950	11	$\frac{850 + 950}{2} = 900$	900 x 11 = 9900
950 – 1050	7	$\frac{950 + 1050}{2} = 1000$	1000 x 7 = 7000
Total	50		39200

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